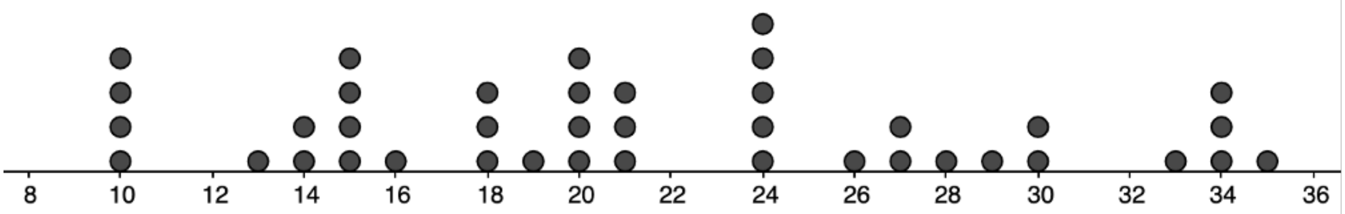
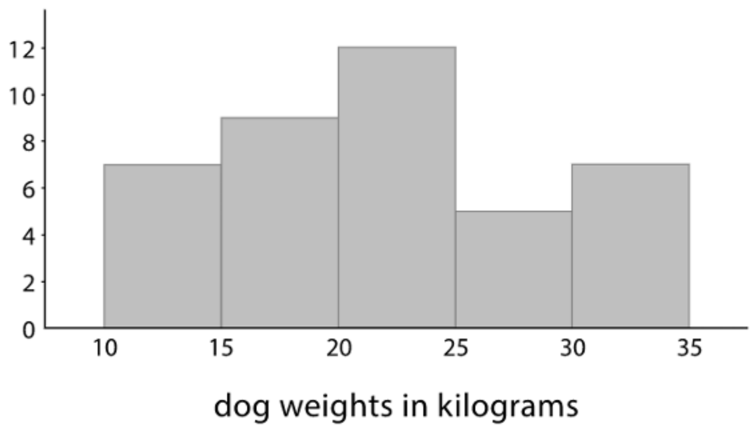
Grade 6
Unit 9
Lesson 8 Practice

Name Answer Key ____________
Date_____________________
Period____________________

Here is a histogram and dot plot showing the weights, in pounds, of 40 dogs at a dog show.





1. Which data display can be used to determine the mean weight?
   Dot plot

2. Find the mean weight. Interpret the mean.
   $[4(10) + 13 + 2(14) + 4(15) + 16 + 3(18) + 19 + +4(20) + 3(21) + 5(24) + 26 + 2(27) + 28 + 29 + 2(30) + 33 + 3(34) + 35] \div 40 = 21.5$

   The average is 21.5 kg.

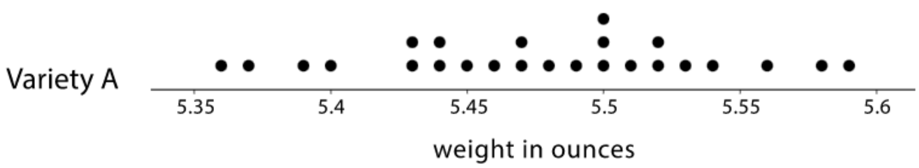
3. Which data display can be used to determine the median weight?
   Dot plot

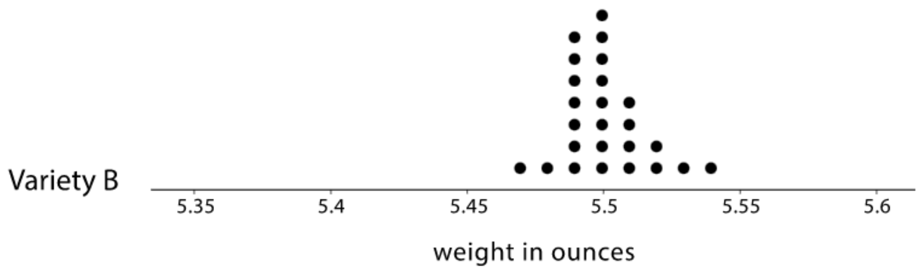
4. Find the median weight. Interpret the median.

   20.5
   The median weight of the dogs is 20.5 kg.

A farmer sells tomatoes in packages of ten. She would like the tomatoes in each package to all be about the same size and close to 5.5 ounces in weight.

The farmer is considering two different tomato varieties: Variety A and Variety B. The dot plots show her data.





5. Describe the shape and distribution of Variety A.
   The data appear symmetric but spread out.

6. Describe the shape and distribution of Variety B.
   The data are clustered and slightly right skewed.

7. What percentage of tomatoes in Variety A were less than 5.49 ounces?
   $\frac{13}{25} = \frac{52}{100}$ 52%

8. What percentage of tomatoes in Variety B were less than 5.49 ounces?
   $\frac{2}{25} = \frac{8}{100}$ 8%

9. What is the range of weights in Variety A and B?

Variety A	Variety B
$5.59 - 5.36 = 0.23$	$5.54 - 5.37 = 0.07$

10. Which tomato variety should the farmer choose? Explain your reasoning.

    The farmer should choose Variety B because the tomatoes are all closer to the same size.